GLOBAL REGULARITY FOR CRITICAL STOCHASTIC SQG WITH FINITE-DIMENSIONAL NONLINEAR NOISE

KAIWEN JIN

Abstract. We prove almost-sure global regularity and pathwise uniqueness for the critical surface quasi-geostrophic equation on the two-dimensional torus with a finite collection of smooth nonlinear state-dependent Itô noise coefficients. For every deterministic smooth datum, globally scalar-Lipschitz coefficients with bounded positive-order derivatives give a global adapted solution continuous in H^3 on every finite time interval, with the integral equation holding in H^1. There is no smallness, mean-zero or affine-noise condition. The proof integrates finite powers of spatial increments. A dissipation bound proportional to the power permits a spatial exponent independent of that power, so a sufficiently large finite power controls a positive Hölder norm. A fractional stochastic truncation argument bounds the amplitude, and a tame H^3 estimate then prevents finite-time blowup.

1. Introduction

The surface quasi-geostrophic equation is a nonlocal active scalar model in geophysical fluid dynamics. Its velocity is recovered from the temperature by Riesz transforms. With order-one dissipation, transport and diffusion act at the same differential order, and an energy estimate alone does not control smooth solutions. Deterministic critical SQG nevertheless has global regularity; Constantin, Tarfulea and Vicol [4], for example, obtain it by estimating spatial increments and propagating a small positive Hölder exponent from an amplitude bound.

We consider the Itô equation

$$\mathrm{d}\theta + \big(R^{\perp}\theta \cdot \nabla\theta + \Lambda\theta\big)\,\mathrm{d}t = \sum_{k=1}^{K} g_k(x,\theta)\,\mathrm{d}W_t^k, \qquad \theta(0) = \theta_0, \tag{1.1}$$

on $\mathbb{T}^2$, where $\Lambda = (-\Delta)^{1/2}$ and $R^{\perp} = (-R_2, R_1)$, with zero Fourier multipliers on constants. The W^k are independent standard Brownian motions on a prescribed filtered probability space satisfying the usual conditions. The datum θ_0 is any deterministic smooth real periodic function. We assume that the finite family $g_k \in C^{\infty}(\mathbb{T}^2 \times \mathbb{R}; \mathbb{R})$ is globally Lipschitz in its scalar argument, uniformly in x, that $g_k(\cdot, 0)$ is bounded, and that

$$\sup_{x,r} |\partial_x^{\beta} \partial_r^{j} g_k(x,r)| < \infty \qquad \text{whenever } |\beta| + j \geq 1. \tag{1.2}$$

Each derivative may have its own bound. In particular the noise can depend nonlinearly on the temperature and smoothly on space.

Local pathwise theory for stochastic fractional SQG is available from Röckner, Zhu and Zhu [6]. Global results at subcritical dissipation are discussed in Agresti and Veraar [1, Theorem 8.22]; their Remark 8.23 identifies the critical endpoint as a remaining question for the general coefficient class there. More recently, Chen and Zhang [3, Section 6] proved critical stochastic

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Contact: yui.ui.siki@gmail.com.

well-posedness on $\mathbb{R}^2$ with additive Brownian forcing. A direct replacement of that additive forcing by $g(x,\theta)$ would require an a priori spatial regularity bound for the composed coefficient, which the energy estimate does not provide. The finite smooth nonlinear class (1.2) gives a concrete critical problem beyond this additive setting. It is the class specified in Problem 478 of the AIM catalogue [7].

Theorem 1.1. *For every datum and finite coefficient family just described, equation (1.1) has a pathwise unique global adapted solution on the prescribed stochastic basis. Almost surely, simultaneously for every finite T,*

$$\theta \in C([0,T];H^3(\mathbb{T}^2)),$$

and, for all $t \in [0,T]$, the integral identity

$$\theta(t) = \theta_0 - \int_0^t \big(R^\perp\theta(s)\cdot\nabla\theta(s) + \Lambda\theta(s)\big)\,\mathrm{d}s + \sum_{k=1}^{K}\int_0^t g_k(\cdot,\theta(s))\,\mathrm{d}W_s^k$$

holds in $H^1(\mathbb{T}^2)$. No smallness or condition on the spatial mean is imposed.

The theorem gives a negative answer to finite-time blowup with positive probability in the full smooth finite-noise class of [7]. In particular, no affine or additive restriction on g_k is introduced. The conclusion concerns this stated class; the broader rough or infinite-dimensional coefficient classes in [1] are not asserted here.

The deterministic increment estimates in [4] suggest propagating a weighted pointwise supremum. For Itô paths, their direct time-differentiation argument is unavailable. We instead integrate a finite power of the weighted increments over space and separation. The scalar convexity remainder retains a factor proportional to that power. It can therefore absorb the transport of the weight with a spatial exponent chosen independently of the power. Taking the power large, but finite, then gives a positive Hölder exponent by a fractional Sobolev estimate. The stochastic trace and bracket are controlled by the same integrated quantity.

Before fixing an amplitude level, a fractional stochastic truncation argument shows that the amplitude is finite up to the possible maximal lifetime. This uses the martingale first-failure method of Hsu, Wang and Wang [5], with the order-one dissipation and the periodic $H^{1/2}\to L^4$ inequality included explicitly. The positive Hölder bound then makes transport absorbable in an H^3 energy estimate. A tame chain rule controls the nonlinear noise using the amplitude, and logarithmic energy bounds exclude a finite maximal lifetime.

Section 2 gives the maximal local solution and the martingale bounds. Section 3 proves the amplitude estimate. Section 4 establishes the integrated increment estimate and completes the continuation argument. We use $\mathbb{T}^2 = (\mathbb{R}/(2\pi\mathbb{Z}))^2$ with ordinary Lebesgue measure and $\widehat{v}(m) = (2\pi)^{-2}\int_{\mathbb{T}^2} v(x)e^{-im\cdot x}\,\mathrm{d}x$. The symbols of Λ and R_j are $|m|$ and $im_j/|m|$ for $m\neq 0$, and zero for $m=0$.

2. Local solutions and stochastic estimates

We use real inhomogeneous Sobolev spaces on the torus. Their norms are

$$\|v\|_{H^s}^2 = (2\pi)^2\sum_{m\in\mathbb{Z}^2}(1+|m|^2)^s|\widehat{v}(m)|^2.$$

The norm in C^γ, $0<\gamma<1$, includes the supremum and the γ-Hölder seminorm. All constants may depend on the fixed coefficient family and the torus. Other dependencies are indicated locally. Constants in estimates at a local stopping time will be independent of its level unless explicitly stated otherwise.

2.1. **The maximal local solution.** The constant mode must remain in the solution space because the noise need not preserve the mean. The abstract local theorem of Röckner, Zhu and Zhu [6, Theorem 3.2 and Definition 3.1] accommodates this directly.

Proposition 2.1. *Under the hypotheses of Theorem 1.1, there is a pathwise unique maximal local solution θ on the prescribed stochastic basis, with lifetime $\tau_* > 0$ almost surely. It satisfies*

$$\theta \in C([0,\tau_*);H^3), \qquad \theta \in L^2_{\mathrm{loc}}([0,\tau_*);H^{7/2}) \quad \text{almost surely.}$$

There are increasing local norm stopping times $\tau_n \uparrow \tau_*$, with deterministic H^3 bounds at each fixed level. If $\tau_* < \infty$, then $\sup_{t<\tau_*}\|\theta(t)\|_{H^3} = \infty$.

Proof. Let P_0 denote orthogonal projection onto constants in $L^2(\mathbb{T}^2)$ and set

$$A = -\Delta + P_0, \qquad f(v) = R^\perp v \cdot \nabla v - P_0 v, \qquad G(v)e_k = g_k(\cdot, v). \tag{2.1}$$

The operator A is positive self-adjoint on full L^2, has domain H^2 and compact inverse, and $A^{1/2} = \Lambda + P_0$. Its spectral Sobolev norms are equivalent to the inhomogeneous norms, including the constant mode. Thus the source equation $\mathrm{d}\theta + (f(\theta) + A^{1/2}\theta)\,\mathrm{d}t = G(\theta)\,\mathrm{d}W$ is precisely (1.1). In the notation of that theorem the regularity index is $s_0 = 3$ and the dissipation exponent is $1/2$.

We verify the drift and noise hypotheses. The periodic product, commutator and Sobolev inequalities in [6, Lemmas 5.1–5.3 and Section 5.2] give, for $s \in [3,4]$ and $u = R^\perp v$,

$$\begin{aligned} |\langle \Lambda^s(u\cdot\nabla v), \Lambda^s v\rangle| &\le C\|v\|_{H^s}\big(\|\nabla u\|_4\|\Lambda^s v\|_4 + \|\Lambda^s u\|_4\|\nabla v\|_4\big) \\ &\le C\|v\|_{H^{3/2}}\|v\|_{H^s}\|v\|_{H^{s+1/2}} \\ &\le \varepsilon\|v\|^2_{H^{s+1/2}} + C_\varepsilon\|v\|^2_{H^3}\|v\|^2_{H^s}. \end{aligned} \tag{2.2}$$

The first line uses transport cancellation and commutator exponents all equal to 4; $H^{1/2} \hookrightarrow L^4$ applies to the differentiated, zero-mean factors. The mean of $u\cdot\nabla v$ is zero, so its exact A-spectral energy equals the homogeneous pairing, while the additional drift in (2.1) contributes $-\|P_0 v\|_2^2$. This proves the source coercivity condition in its explicitly stated equivalent form. For $s = 2, 3$, the same product estimate gives

$$\|f(v)\|_{H^s} \le C(1 + \|v\|_{H^3})\|v\|_{H^{s+3/2}}.$$

These bounds also give the required continuous drift maps and smooth-test pairings. For $q = v_1 - v_2$, divergence-free transport cancels and leaves $\langle q, R^\perp q\cdot\nabla v_1\rangle$ and the bounded mean term. Consequently the local L^2 monotonicity condition follows from $\|\nabla v_1\|_\infty \le C\|v_1\|_{H^3}$ and the Riesz L^2 bound.

The coefficient assumptions imply

$$\sum_k \|g_k(\cdot, v)\|^2_{H^s} \le C(\|v\|_{H^3})(1 + \|v\|^2_{H^s}), \qquad 3 \le s \le 4, \tag{2.3}$$

with a locally bounded coefficient, as well as the global L^2 Lipschitz bound for G. Here is a verification of (2.3) allowing full spatial and scalar dependence. Write $s = 3 + \sigma$, $0 \le \sigma \le 1$. Every third spatial derivative of $g(x, v(x))$ has the form $g_r(x,v)D^3 v + F$, where F is a sum of bounded-coefficient products of DvD^2v, $(Dv)^3$, D^2v, $(Dv)^2$, Dv and a bounded function. On an H^3 ball, $Dv \in L^\infty$, $D^2v \in L^4$ and $D^3v \in L^2$. Differentiating F once, including the product $(D^2v)^2$, therefore shows $\|F\|_{H^1} \le C(\|v\|_{H^3})$. Also $a(x) = g_r(x, v(x))$ has H^3 norm bounded on that ball by the integer chain rule. Fourier convolution gives, uniformly for $0 \le \sigma \le 1$,

$$\|az\|_{H^\sigma} \le C\Big(\sum_m (1 + |m|)^\sigma |\widehat{a}(m)|\Big)\|z\|_{H^\sigma} \le C\|a\|_{H^3}\|z\|_{H^\sigma}.$$

The last sum is controlled by Cauchy–Schwarz since $\sum_m (1+|m|)^{-6+2\sigma} < \infty$ uniformly in this interval. This proves (2.3); lower derivatives and the L^2 term follow from the same chain rule

and linear growth. Finiteness of the family removes any summability issue. The L^2 Lipschitz bound follows directly from scalar Lipschitz continuity.

All hypotheses of the cited abstract theorem now hold on the original basis. The datum is deterministic and has finite H^3 norm. The theorem and its maximal-solution definition yield the asserted lifetime and pathwise uniqueness. Intersecting its local exhaustion with the first H^3 level-n hit, if necessary, gives the indicated norm stops without changing the lifetime. □

The regularity in Proposition 2.1 justifies the calculations below at each fixed local stop. In particular $H^3(\mathbb{T}^2) \hookrightarrow C^{1,\beta}$ for every $0 < \beta < 1$, for both θ and u. This follows directly from Cauchy–Schwarz in the Fourier sum with weight $(1+|m|^2)^{-3}|m|^{2+2\beta}$. No higher local smoothness is required.

2.2. **Martingale bounds.** We use the following two forms of the continuous martingale estimate.

Lemma 2.2. *If M is a continuous local martingale, $a, b > 0$ and $S < \infty$, then*

$$\mathbb{P}\left\{\sup_{0\le s\le t\le S}(M_t - M_s) > a, [M]_S \le b\right\} \le C\exp(-ca^2/b) \tag{2.4}$$

for universal positive constants. Suppose nonnegative continuous semimartingales X_n have a common deterministic initial value and, before prescribed stops,

$$\mathrm{d}X_n \le C(1+X_n)\,\mathrm{d}t + \mathrm{d}M_n, \qquad \mathrm{d}[M_n]_t \le C(1+X_n)^2\,\mathrm{d}t, \tag{2.5}$$

with C independent of n. On every fixed horizon, the suprema of the stopped $\log(1+X_n)$ have a common tail tending to zero, and all finite moments of the stopped $\sup X_n$ are bounded uniformly in n.

Proof. Estimate (2.4) is the continuous local martingale oscillation bound of Hsu, Wang and Wang [5, Lemma 3.1, arXiv version 3]; only unspecified universal constants are needed here. Applying Itô's formula to (2.5) gives, on $[0, S]$,

$$\log(1+X_n(t)) \le \log(1+X_n(0)) + CS + L_n(t), \qquad [L_n]_S \le CS.$$

The second derivative term has nonpositive sign. Formula (2.4), with the initial time included, gives a Gaussian upper tail for $\sup L_n$. Its exponential moments of every finite order are finite, proving both assertions. A zero bracket gives the same statements directly. Auxiliary martingale stops can first be inserted and then removed. □

We apply finite-dimensional or scalar Itô calculus before taking spatial limits; see [2, Theorem 2.4]. The finite family and the local H^3 bounds ensure the required integrability. Stochastic integrals commute with the spatial integrals below by approximation with simple functions and the stopped L^2 isometry. For an indicator before a stopping time one can use the predictable version $1_{t\le\tau_n}$, which gives the same Brownian integral as $1_{t<\tau_n}$.

3. An amplitude bound before the maximal lifetime

The coefficient assumptions give

$$\sum_k |g_k(x,r)|^2 \le C(1+r^2). \tag{3.1}$$

We first prove an amplitude estimate using only this growth bound and the divergence-free property of the drift. In particular, its constants do not involve an assumed amplitude bound.

Proposition 3.1. *For $0 < \varepsilon < S < \infty$ and every finite $q \geq 1$,*

$$\sup_n \mathbb{E}\left[\sup_{\varepsilon \leq t \leq S,\ t<\tau_n} \|\theta(t)\|_\infty^q\right] < \infty, \tag{3.2}$$

where an empty supremum is zero. Almost surely, for every finite S,

$$\sup_{0 \leq t < S \wedge \tau_*} \|\theta(t)\|_\infty < \infty. \tag{3.3}$$

Proof. The L^2 Itô identity, transport cancellation and (3.1) imply

$$\sup_n \mathbb{E} \sup_{t \leq S \wedge \tau_n} (1 + \|\theta(t)\|_2^2)^q < \infty \quad (S < \infty, q < \infty). \tag{3.4}$$

Indeed the drift of $X = \|\theta\|_2^2$ is at most $C(1+X)$ after dropping $2\|\Lambda^{1/2}\theta\|_2^2$, and the energy martingale has bracket at most $C(1+X)^2\,\mathrm{d}t$. Lemma 2.2 applies.

Fix n. Set $\widehat{\theta}_n(t) = \theta(t)$ for $t < \tau_n$ and $\widehat{\theta}_n(t) = 0$ afterwards. For $a \geq 0$ the energy of $z = (\widehat{\theta}_n - a)_+$ has a downward jump at the killing time. Before that time its truncation inequality is

$$\mathrm{d}\|z\|_2^2 + 2\|\Lambda^{1/2} z\|_2^2\,\mathrm{d}t \leq C \int_{\theta > a} (1+\theta^2)\,\mathrm{d}x\,\mathrm{d}t + 2\sum_j \left(\int z g_j(x,\theta)\,\mathrm{d}x\right) \mathrm{d}W^j. \tag{3.5}$$

Every integrand is understood to vanish after killing. The transport term is $\int u \cdot \nabla(z^2)/2 = 0$. For the fractional term, integrate

$$(r-s)\big((r-a)_+ - (s-a)_+\big) \geq \big|(r-a)_+ - (s-a)_+\big|^2$$

against the symmetric positive kernel of Λ.

To justify the nonsmooth truncation, take smooth convex functions ϕ_η vanishing on $(-\infty, a]$, with $0 \leq \phi_\eta'' \leq 2$ and $\phi_\eta' \to 2(r-a)_+$ uniformly on bounded scalar intervals. Pointwise Itô's formula, followed by spatial integration, applies at a fixed H^3 stop. The fractional bilinear forms converge by the local Lipschitz bound, the trace terms are dominated by $C(1+\theta^2)$, and the stochastic terms converge by the stopped L^2 isometry. This proves (3.5). Its integrated version holds at all time pairs for a continuous martingale version. Killing only subtracts the left truncation energy. After killing, $z = 0$, so its dissipation also vanishes on the rest of a deterministic interval.

We carry out the fractional version of the probability iteration in [5, Sections 2–3]. Fix $T > 0$, $a \geq 1$, and let

$$I_k = [(1-2^{-k})T, 2T], \quad a_k = a(1-2^{-k}), \quad z_k = (\widehat{\theta}_n - a_k)_+, \quad U_k = \left(\int_{I_k} \|z_k(t)\|_2^4\,\mathrm{d}t\right)^{1/2}.$$

For $k \geq 1$, on $\{z_k > 0\}$ one has $z_{k-1} \geq a2^{-k}$ and $1 + |\theta| \leq C2^k z_{k-1}$. Let X_{k-1}^* be the supremum of increments over time pairs in I_{k-1} of the continuous martingale

$$X(t) = \sum_j \int_0^t \left(\int z_k g_j(x,\theta)\,\mathrm{d}x\right) \mathrm{d}W^j.$$

Averaging the initial L^2 energy in $I_{k-1} \setminus I_k$, whose length is $2^{-k}T$, and then integrating (3.5) gives

$$\sup_{I_k} \|z_k\|_2^2 + \int_{I_k} \|\Lambda^{1/2} z_k\|_2^2\,\mathrm{d}t \leq C_T^k (U_{k-1} + X_{k-1}^*). \tag{3.6}$$

For this averaging, $\int_{I_{k-1}} \|z_{k-1}\|_2^2 \,\mathrm{d}t \le (2T)^{1/2} U_{k-1}$. The trace drift is bounded by C^k times that integral, and every martingale difference is bounded by X^*_{k-1}. Thus the selected initial time is chosen pathwise; it need not be a stopping time.

The inhomogeneous periodic Sobolev inequality is

$$\|z\|_4^2 \le C\big(\|\Lambda^{1/2} z\|_2^2 + \|z\|_2^2\big). \tag{3.7}$$

It follows from the homogeneous $H^{1/2}$ inequality after subtracting the mean. The extra L^2 term is bounded in time by $(2T)^{1/2} U_{k-1}$. Chebyshev's inequality gives $|\{z_k(t) > 0\}| \le (2^k/a)^2 \|z_{k-1}(t)\|_2^2$. Spatial Hölder, followed by time Hölder, therefore yields

$$U_k \le (2^k/a)^{2/5} \|z_k\|^2_{L^5(I_k; L^{5/2})} U_{k-1}^{1/5}.$$

The interpolation $\|z\|_{L_t^5 L_x^{5/2}} \le \|z\|^{3/5}_{L_t^\infty L_x^2} \|z\|^{2/5}_{L_t^2 L_x^4}$, together with (3.6)–(3.7), proves

$$U_k \le C_T^k a^{-2\delta} (U_{k-1} + X^*_{k-1}) U_{k-1}^{\delta}, \qquad \delta = \frac{1}{5}. \tag{3.8}$$

This value uses the actual order-one fractional dissipation in dimension two.

By (3.1) and Cauchy–Schwarz in space, the bracket of X over I_{k-1} is bounded by $C^k U_{k-1}^2$. Start this martingale at the deterministic left endpoint of that interval. Lemma 2.2 gives

$$\mathbb{P}\{X^*_{k-1} > \ell b,\ U_{k-1} \le b\} \le C \exp(-\ell^2/C^k), \qquad b, \ell > 0. \tag{3.9}$$

Only the implication from $U_{k-1} \le b$ to the bracket bound is used.

Here is the threshold choice for (3.8). Enlarge $B \ge 2$ so the bracket constants are at most B^k, and set

$$b_k = a^2 \lambda^k / \varrho^2, \qquad \ell_k = (2B)^{k/2} \varrho^{\delta}.$$

Choose $\lambda \in (0,1)$ so that $C_T (2B)^{1/2} \lambda^{\delta} \le 1/4$, and then choose $\varrho \ge \varrho_0(T)$ sufficiently large. If $U_{k-1} \le b_{k-1}$ and $X^*_{k-1} \le \ell_k b_{k-1}$, division of (3.8) by b_k bounds the result by

$$C_T^k (1 + \ell_k) \lambda^{\delta(k-1)-1} \varrho^{-2\delta} \le C \lambda^{-1-\delta} \varrho^{-\delta} \le 1.$$

On $U_0 \le a^2/\varrho^2$, the first failed level must therefore lie in an event in (3.9). Their probabilities sum to at most $C \sum_{k \ge 1} \exp(-c 2^k \varrho^{2\delta})$. If none fails, $U_k \to 0$, which forces $(\widehat{\theta}_n - a)_+ = 0$ almost everywhere on $[T, 2T] \times \mathbb{T}^2$. Hence

$$\mathbb{P}\left\{ \|\widehat{\theta}_n^+\|_{L^\infty([T,2T]\times\mathbb{T}^2)} > a,\ \varrho U_0^{1/2} \le a \right\} \le C \exp(-c\varrho^{\delta}). \tag{3.10}$$

All these constants are independent of n.

Take $\varrho = \sqrt{a}$ for sufficiently large a. By (3.4), U_0 has moments of every finite order, uniformly in n. The probability in the complementary event is $\mathbb{P}\{U_0 > a\}$, so (3.10) yields amplitude moments of every finite order. Apply the same calculation to $-\theta$. Continuity before killing turns the essential bounds into pointwise time and space suprema. A finite covering by positive-time intervals proves (3.2).

Monotone exhaustion of τ_n gives finite amplitude on $[\varepsilon, S \wedge \tau_*)$ almost surely, simultaneously for rational $\varepsilon > 0$ and integer S. Since $\tau_* > 0$, each sample admits an integer m with $1/m < \tau_*$. Local H^3 continuity bounds the solution on $[0, 1/m]$; the positive-time estimate bounds the remaining interval. This proves (3.3). □

4. Spatial regularity and global continuation

We now fix an integer amplitude level $M \ge 1$. All estimates in this section are initially stopped both at τ_n and at $\sigma_M = \inf\{t < \tau_* : \|\theta(t)\|_\infty \ge M\}$, with the infimum of the empty set equal to infinity. The exponents below are chosen deterministically for that M.

4.1. **Dissipation of finite powers of increments.** Write $u = R^\perp\theta$, $w(x,h) = \delta_h\theta(x) = \theta(x+h) - \theta(x)$, using periodic extension in $h \in \mathbb{R}^2$. Then

$$\mathrm{d}w + (u(x)\cdot\nabla_x + \delta_h u(x)\cdot\nabla_h + \Lambda_x)w\,\mathrm{d}t = \sum_k \delta_h g_k\,\mathrm{d}W^k, \tag{4.1}$$

where $\delta_h g_k = g_k(x+h,\theta(x+h)) - g_k(x,\theta(x))$. Both transport fields are divergence free in their integration variables. The bounded scalar and spatial derivatives imply

$$\left(\sum_k |\delta_h g_k|^2\right)^{1/2} \le L_0|w| + B_0|h|. \tag{4.2}$$

The constants are independent of derivatives of θ.

The following elementary scalar inequality preserves the dependence on p needed to choose a spatial exponent before selecting a large finite p.

Lemma 4.1. *For real a, b and $p \ge 2$,*

$$|b|^p - |a|^p - p|a|^{p-2}a(b-a) \ge \frac{p}{2}|a|^{p-2}(a-b)^2. \tag{4.3}$$

Consequently, with

$$D[w](x) = \frac{1}{2\pi}\int_{\mathbb{R}^2} \frac{|w(x) - w(x+y)|^2}{|y|^3}\,\mathrm{d}y,$$

one has

$$p|w|^{p-2}w\Lambda w - \Lambda(|w|^p) \ge \frac{p}{2}|w|^{p-2}D[w]. \tag{4.4}$$

Proof. For $a \neq 0$ divide by $|a|^p$ and put $t = b/a$. If $t \ge 0$, then

$$t^p - 1 - p(t-1) - (p-1)(t-1)^2 = t\big(t^{p-1} - (p-1)t + p - 2\big) \ge 0$$

by convexity, and $p - 1 \ge p/2$. If $t = -s \le 0$, convexity of $q \mapsto q^{p/2}$ gives $s^p \ge (p/2)s^2 + 1 - p/2$ and hence

$$s^p + ps + p - 1 \ge (p/2)(s+1)^2.$$

The case $a = 0$ follows directly; at $p = 2$ interpret $|a|^0 = 1$, and the inequality is an equality. Integrate (4.3) against the positive kernel of Λ to obtain (4.4). □

We recall the spatial estimates of Constantin, Tarfulea and Vicol [4, (4.15)–(4.25), August 2, 2013 author version]:

$$D[w] \ge c_0\frac{|w|^3}{M|h|}, \qquad |\delta_h u| \le C_0\left(\sqrt{rD[w]} + \frac{M|h|}{r}\right), \qquad r \ge 4|h|. \tag{4.5}$$

Their whole-$\mathbb{R}^2$ periodic-kernel proof also makes the role of the mean and of large splitting radii explicit. For a radial cutoff η equal to one on $[0,1]$ and zero beyond 2, put $F_r(y) = (1 - \eta(|y|/r))|y|^{-3}$. Expanding the square in D and shifting the finite difference in the cross term gives

$$D[w] \ge c|w|^2/r - C|w|M|h|/r^2, \qquad \|\delta_{-h}F_r\|_1 \le C|h|/r^2.$$

Selecting $r = c_{\mathrm{r}}M|h|/|w|$, with a sufficiently large universal c_{r}, proves the lower bound. For the velocity split the Riesz kernel $y^\perp/(2\pi|y|^3)$ at radius r. The radial inner kernel has zero integral. Subtracting $w(x)$ and applying Cauchy–Schwarz gives $C\sqrt{rD[w]}$. In the outer part, shift δ_h to the kernel; its derivative has L^1 norm at most C/r, giving $CM|h|/r$. The far scalar kernel is integrable, and the shifted Riesz kernel is integrable. With a large outer cutoff of radius L, the shift error is $O(M|h|/L)$ and disappears as $L \to \infty$. These bounds hold for every $r \ge 4|h|$,

including radii beyond a fundamental period. Constants cancel in the increments, so no zero-mean condition is involved. Vanishing w is treated separately before division.

Take now $r = c_{\mathrm{r}} M|h|/|w|$ with a fixed $c_{\mathrm{r}} \geq 8$. Since $|w| \leq 2M$, this radius is admissible. Young's inequality gives

$$\alpha \frac{|\delta_h u|}{|h|}|w|^2 \leq \frac{1}{8} D[w] + C\left(\alpha^2 c_{\mathrm{r}} M + \frac{\alpha}{c_{\mathrm{r}}}\right)\frac{|w|^3}{|h|}.$$

The lower bound in (4.5) shows that a sufficiently small universal $c_* > 0$ allows

$$0 < \alpha < \min\{1/4, c_*/M\}, \qquad \alpha \frac{|\delta_h u|}{|h|}|w|^2 \leq \frac{1}{4} D[w]. \tag{4.6}$$

The choice of α is independent of p. Multiplication by $p|w|^{p-2}$ lets (4.4) absorb the derivative of the singular increment weight for any finite p.

4.2. **An integrated increment estimate.** Choose a nonnegative $\chi \in C_c^\infty(\mathbb{R}^2)$ equal to one near zero, with support in a small ball. For a fixed even integer $p > 8/\alpha$ define

$$Y(t) = \int_{\mathbb{T}^2}\int_{\mathbb{R}^2} \chi(h)\frac{|\delta_h \theta(x,t)|^p}{|h|^{\alpha p}}\,\mathrm{d}h\,\mathrm{d}x. \tag{4.7}$$

The measure in h is ordinary Lebesgue measure. Thus Y is a truncated fractional Sobolev seminorm of order $s = \alpha - 2/p$, and the Hölder exponent it controls in dimension two is

$$\gamma = \alpha - 4/p > 0. \tag{4.8}$$

Proposition 4.2. *For each fixed M and the deterministic exponents in (4.6)–(4.8), the pre-lifetime C^γ norm is almost surely bounded on every finite horizon before the amplitude-M stop. The bounds obtained by local stopping are uniform in n.*

Proof. Apply Itô's formula to $|w|^p$ in (4.1) and integrate against $\chi(h)|h|^{-\alpha p}$. Spatial transport integrates to zero, as does $\int \Lambda_x(|w|^p)\,\mathrm{d}x$. The weight's transport derivative contributes

$$-\alpha p \int \chi(h)|w|^p |h|^{-\alpha p - 2}\delta_h u \cdot h\,\mathrm{d}h\,\mathrm{d}x,$$

which is absorbed using (4.4) and (4.6). The derivative of χ is supported away from zero and has bounded contribution depending on M, p, since $|w| \leq 2M$ and $\|\delta_h u\|_1 \leq C\|\theta\|_2 \leq CM$.

Let $J = \int \chi(h)|h|^{p(1-\alpha)}\,\mathrm{d}h\,\mathrm{d}x < \infty$. By (4.2) and Hölder, the Itô trace correction is at most

$$C_p\left(Y + Y^{(p-2)/p} J^{2/p}\right) \leq C_p(1+Y),$$

and the Euclidean norm of the integrated martingale coefficient is at most

$$pL_0 Y + pB_0 Y^{(p-1)/p} J^{1/p} \leq C_p(1+Y).$$

We have therefore proved

$$\mathrm{d}Y \leq C_{M,p}(1+Y)\,\mathrm{d}t + \mathrm{d}\mathcal{N}, \qquad \mathrm{d}[\mathcal{N}]_t \leq C_p(1+Y)^2\,\mathrm{d}t. \tag{4.9}$$

For completeness, all integrations are first carried out with $|h| > \epsilon$ at a fixed local H^3 stop. Then $w = O(|h|)$ and $\delta_h u = O(|h|)$, so the inner boundary flux is $O(\epsilon^{2+p(1-\alpha)})$. Choose $1/2 < \beta < 1$. Local $C^{1,\beta}$ regularity gives

$$|\delta_y w| \leq C_n |h| \min\{|y|^\beta, 1\}, \qquad D[w] \leq C_n |h|^2.$$

Here the near-y integral is bounded by $\int_0^1 r^{2\beta-2}\,\mathrm{d}r < \infty$. After spatial cancellation, the remaining drift and dissipation have integrable bounds of order $|h|^{p(1-\alpha)}$; the noise terms have the same integrable powers. Dominated convergence and the stopped stochastic isometry remove ϵ before any uniform limit in n is taken. Thus (4.9) holds for Y itself at the local and amplitude stops.

The datum is smooth, so $Y(0) < \infty$. Lemma 2.2 supplies a tail tending to zero for its stopped supremum uniformly in n. The suprema on the increasing stopped intervals are increasing, so their limit is finite almost surely.

We spell out the embedding to keep track of the loss in (4.8). For a sufficiently small torus ball B_r, all pairwise differences lie where $\chi = 1$. Jensen's inequality and (4.7) give

$$\fint_{B_r} |\theta - \theta_{B_r}|^p \le |B_r|^{-2} \int_{B_r}\int_{B_r} |\theta(x) - \theta(y)|^p \,\mathrm{d}x\,\mathrm{d}y \le C r^{\alpha p - 4} Y = C r^{p\gamma} Y.$$

Differences of averages on concentric radii $r, r/2$ are bounded by $C r^\gamma Y^{1/p}$. Sum the geometric series and use local continuity to obtain $|\theta(x) - \theta_{B_r(x)}| \le C r^\gamma Y^{1/p}$. For $|x - y| = r$, compare these averages in a common ball of radius $3r$. For larger distances use the amplitude bound. Consequently

$$\|\theta\|_{C^\gamma} \le C_{\alpha,p,\chi}(M + Y^{1/p}), \tag{4.10}$$

which proves the proposition. □

4.3. **The high-regularity estimate.** We need a continuation bound that uses only a positive Hölder exponent and the amplitude. Fix a periodic inhomogeneous Littlewood–Paley decomposition $(\Delta_j)_{j\ge -1}$ and put $E_s = \sum_{j\ge -1} 2^{2sj}\|\Delta_j\theta\|_2^2 \asymp \|\theta\|_{H^s}^2$. The low block includes the constant mode.

Lemma 4.3. *For smooth real periodic θ, $u = R^\perp\theta$, $0 < \gamma < 1$ and $s \ge 3$,*

$$\left|\sum_j 2^{2sj}\langle \Delta_j(u\cdot\nabla\theta), \Delta_j\theta\rangle\right| \le C_{s,\gamma}\|\theta\|_{C^\gamma}\|\theta\|_{H^{s+(1-\gamma)/2}}^2. \tag{4.11}$$

For $\|\theta\|_\infty \le M$, one also has

$$\sum_k \|g_k(\cdot,\theta)\|_{H^3}^2 \le C_M(1 + \|\theta\|_{H^3}^2). \tag{4.12}$$

Proof. Let $\mathcal{K} = \|\theta\|_{C^\gamma}$, and write $\theta_j = \Delta_j\theta$ and $S_{j-1} = \sum_{\ell\le j-2}\Delta_\ell$. The periodic block kernels are periodizations of Schwartz kernels; their L^1 norms are bounded and their first moments are $O(2^{-j})$. Their zero average, also for the smooth annular multiplier $R\Delta_j$, gives

$$\|\theta_j\|_\infty + \|u_j\|_\infty \le C\mathcal{K}2^{-\gamma j}, \qquad \|\nabla S_{j-1}u\|_\infty + \|\nabla S_{j-1}\theta\|_\infty \le C_\gamma \mathcal{K} 2^{(1-\gamma)j}.$$

The finitely many low modes satisfy the corresponding bound directly.

Decompose the product into low–high, high–low and neighboring high–high terms. In low–high interactions $|k - j| \le C$. Commuting Δ_j past $S_{k-1}u\cdot\nabla$ costs

$$C2^{-j}\|\nabla S_{k-1}u\|_\infty 2^k\|\theta_k\|_2.$$

The remaining principal transport is $S_{j-1}u\cdot\nabla\theta_j$, whose energy pairing is zero. Its difference from $S_{k-1}u$ costs $C\mathcal{K}2^{-\gamma j}$ times the derivative of the neighboring block. High–low terms cost $\|u_k\|_2\|\nabla S_{k-1}\theta\|_\infty$ for the same neighboring indices. These terms are bounded by $C\mathcal{K}\|\theta\|_{H^{s+(1-\gamma)/2}}^2$.

For high–high terms use $u\cdot\nabla\theta = \nabla\cdot(u\theta)$. The block norm is bounded by

$$C\mathcal{K}2^j \sum_{k\ge j-C} 2^{-\gamma k}\|\theta_k\|_2.$$

Set $\sigma = s + (1-\gamma)/2$ and $b_j = 2^{\sigma j}\|\theta_j\|_2$. The weighted pairings sum to at most

$$C\mathcal{K}\sum_{j\ge -1}\sum_{k\ge j-C} 2^{-(s+(1+\gamma)/2)(k-j)} b_j b_k \le C\mathcal{K}\sum_j b_j^2.$$

The convolution is summable. The low block costs at most $C\mathcal{K}E_s$. This proves (4.11); the proof also bounds the sum of absolute block pairings after the indicated cancellations.

For (4.12), the third-derivative chain rule contains only the nonlinear highest-order products $D\theta D^2\theta$ and $(D\theta)^3$. Let $\mathfrak{a} = \|D\theta\|_6$, $\mathfrak{b} = \|D^2\theta\|_3$ and $\mathfrak{c} = \|D^3\theta\|_2$. Periodic integration by parts gives

$$\mathfrak{a}^6 \le CM\mathfrak{a}^4\mathfrak{b}, \qquad \mathfrak{b}^3 \le C\mathfrak{a}\mathfrak{b}\mathfrak{c}.$$

For the first inequality integrate $\operatorname{div}(|D\theta|^4 D\theta)$ against θ. For the second integrate one derivative in $\sum_{i,j}\int \theta_{ij}^2 |D^2\theta|$, and use $1/6 + 1/3 + 1/2 = 1$; a smooth regularization of the Hessian length justifies its value at zero. Unless one of the derivative norms vanishes, in which case the estimates are immediate, these inequalities imply

$$\mathfrak{a} \le CM^{2/3}\mathfrak{c}^{1/3}, \qquad \mathfrak{b} \le CM^{1/3}\mathfrak{c}^{2/3}.$$

Thus $\|D\theta D^2\theta\|_2 \le \mathfrak{a}\mathfrak{b} \le C_M\mathfrak{c}$ and $\|(D\theta)^3\|_2 = \mathfrak{a}^3 \le C_M\mathfrak{c}$. All mixed spatial coefficient derivatives are bounded by assumption; lower products obey the same bounds and finite-volume Hölder. Finally $\|g_k(\cdot,\theta)\|_2 \le C(1+M)$, proving (4.12). □

Apply Itô's formula to E_3, first summing only finitely many whole dyadic blocks. Each such block has finite Fourier support on the torus. At a local H^3 stop, the nonlinear pairings converge absolutely in time by the preceding proof and $\theta \in L_t^2 H^{7/2}$; the dissipation converges monotonically. The trace and martingale terms converge by (4.12) and the stopped isometry. This proves the energy formula for E_3 without assuming an H^3-valued drift.

The dissipation bounds the high-frequency $H^{7/2}$ norm. The finitely many low modes cost CE_3, including the nondissipated constant mode. Interpolation and Young's inequality give

$$C\mathcal{K}\|\theta\|_{H^{3+(1-\gamma)/2}}^2 \le \varepsilon\|\theta\|_{H^{7/2}}^2 + C_{\gamma,\varepsilon}\mathcal{K}^{1/\gamma}\|\theta\|_{H^3}^2.$$

Consequently, before the amplitude-M and C^γ level-R stops,

$$\mathrm{d}E_3 \le C_{M,R,\gamma}(1+E_3)\,\mathrm{d}t + \mathrm{d}\mathcal{M}, \qquad \mathrm{d}[\mathcal{M}]_t \le C_M(1+E_3)^2\,\mathrm{d}t. \tag{4.13}$$

For the bracket use Cauchy–Schwarz in the dyadic Hilbert norm and (4.12). Lemma 2.2 now bounds the stopped $\sup E_3$ by tails independent of the local level n.

4.4. **Exclusion of a finite lifetime.**

Proof of Theorem 1.1. For each integer $M \ge 1$ fix deterministic α_M satisfying (4.6), and then an even integer $p_M > 8/\alpha_M$. Put $\gamma_M = \alpha_M - 4/p_M > 0$. Proposition 4.2 yields a finite pre-lifetime C^{γ_M} supremum at the amplitude-M stop on every finite horizon. The uniform local-stop tail and monotone exhaustion justify this assertion. Take a common probability-one event over integers M, finite integer horizons, and the later Hölder levels R in (4.13).

On a sample with finite τ_*, Proposition 3.1 supplies a finite amplitude supremum up to that lifetime. Some integer M strictly exceeds it. For this deterministic family member, the Hölder supremum is finite, so some integer R strictly exceeds it. Neither of these two stops is then reached before τ_*. The uniform H^3 tail in (4.13), exhausted monotonically in n, gives a finite pre-lifetime H^3 supremum. This contradicts Proposition 2.1. Hence $\mathbb{P}\{\tau_* < \infty\} = 0$.

Every exponent was fixed before its Itô estimate. The sample-dependent choice of one integer M, and then R, occurs only after the countable intersection and introduces no anticipating stochastic integrand.

On every finite interval the patched local solution is continuous in H^3 and adapted to the original basis. Local pathwise uniqueness gives global pathwise uniqueness by increasing common norm stops. Its drift is continuous in H^2, since $H^2(\mathbb{T}^2)$ is an algebra. Its noise is predictable and locally square integrable in H^3, so its stochastic integral has a continuous H^3 version [2, Proposition 2.1]. The local weak identity holds on a countable Fourier basis and rational times.

Both sides are continuous in H^2, so Fourier completeness and time continuity give the integral identity at every time in H^2, hence in H^1. This proves all the assertions of the theorem. □

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